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Evaluation of Quality of Life/Life Satisfaction in Women with Breast Cancer in Complementary and Conventional Care

Marianne Carlsson, Maria Arman, Marie Backman, Ursula Flatters, Thomas Hatschek and Elisabeth Hamrin

From the Department of Public Health and Caring Sciences, Section of Caring Sciences, Uppsala University, Sweden (M. Carlsson), the Department of Caring Sciences, Åbo Akademi University, Vasa, Finland and Blekinge Institute of Technology, Karlskrona, Sweden (M. Arman), The Swedish Red Cross University College of Nursing, Stockholm, Sweden (M. Backman), The Vidar Clinic, Järna, Sweden (U. Flatters), the Department of Oncology (Radiumhemmet), Karolinska Hospital and Institute, Stockholm, Sweden (T. Hatschek) and the Department of Medicine and Care, Divisions of Clinical Pharmacology and Nursing Science, Faculty of Health Sciences, University of Linköping, Sweden (E. Hamrin)

Correspondence to: Marianne Carlsson, Uppsala University, Department of Public Health and Caring Sciences, Section of Caring Sciences, Uppsala Science Park, SE-751 83 Uppsala, Sweden. Tel: +46 18 471 3482. Fax: +46 18 471 3490. E-mail: Marianne.Carlsson@pubcare.uu.se

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The aim was to study the perceived quality of life/life satisfaction in a sample of women with breast cancer who were treated in a hospital with alternative/complementary care and the same variables in individually matched patients who received only conventional medical treatment. A non-randomized controlled trial design with repeated measurements was used. Sixty women with breast cancer treated with anthroposophic medicine (ABCW) and 60 with conventional medicine (CBCW) were included and 36 matched pairs took part on all occasions. The quality of life was measured by the EORTC QLQ-C30 and the Life Satisfaction Questionnaire (LSQ). The comparisons were calculated as effect sizes (ES). The women in the ABCW group reported small or moderate effects, expressed as ES, on their quality of life/life satisfaction compared to their matched "twins" in the CBCW group at the 1-year follow-up in 15 out of 21 scales/factors. It was concluded that the women who had chosen anthroposophic care increased their perceived quality of life/life satisfaction according to the methodology of the study.

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Between 16% and 40% of cancer patients have reported the use of alternative/complementary therapies, women more often than men (1–3). A study by Burnstein et al. (4) showed that 38% of a sample of women with breast cancer had used complementary medicine during the previous 12 months.

Astin (5) asked 1,035 randomly selected individuals about their use of alternative medicine during the previous year. The results showed that users of alternative medicine compared to non-users tended to be better educated and had a more holistic orientation towards health, were more likely to have had some type of transformational experience that had changed their worldview in some significant way, and were also likely to report poorer health status.

The principles underlying anthroposophic medicine were developed by Rudolf Steiner (1861–1925), the founder of anthroposophy, and the medical doctor Ita Wegman (1876–1943), and were presented in 1925 in a book entitled *Fundamentals of Therapy, an Extension of the Art of Healing through Spiritual Knowledge*. The name anthroposophy is derived from Greek: *anthropos* (human) and *sophia* (wisdom). According to Evans and Rodger (6), the idea is to develop spiritual wisdom through human self-knowledge. In anthroposophic medicine, natural scientific methods used in conventional medicine are not contraindicated. In order to reach true healing, however, the patient must be given the opportunity to mature through the complementary use of anthroposophic therapies. Anthroposophic care is characterized by a holistic outlook on human beings and uses natural products. The concept "complementary" signifies that conventional therapy can also be part of the treatment.

The only anthroposophic hospital in Scandinavia is situated in the County of Stockholm in Sweden. It is a

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small hospital (74 beds), and about 40% of the patients suffer from cancer.

The aim of the present study was to examine over 1 year the perceived quality of life/life satisfaction in a sample of women with breast cancer who were treated in this hospital with alternative/complementary care and to study the same variables in individually matched patients receiving only conventional medical treatment.

MATERIAL AND METHODS

The present study is part of a major clinical study of the outcome of complementary medicine, in this case anthroposophic medicine, for women with breast cancer. By matching procedures, women with breast cancer who did not receive complementary care were also studied, as was the relationship between different outcome measurements (7). The following outcome measures were used in the main study: (a) perceived quality of life and coping measured by questionnaires (8–10) and tape-recorded interviews; (b) medical investigation; and (c) an immunological test package. The design was based on prestudies performed in 1988–1994 (11).

Patients

The subjects were women diagnosed with breast cancer who chose the anthroposophic hospital for treatment, were 75 years old or younger, able to communicate in Swedish, lived in the County of Stockholm, and expected to survive at least 6 months. In all, 68 women were asked and 60 of those agreed to participate. Those 60 women who met the inclusion criteria were admitted to the anthroposophic hospital from November 1995 to January 1999 (ABCW = women with breast cancer, anthroposophic therapy). Each woman was matched individually with a woman from an oncology outpatient department in the southern Stockholm area who received conventional medical cancer treatment (CBCW = women with breast cancer, conventional therapy).

The matching criteria were based on the following principles: (a) stage of disease at entry—advanced inoperable tumor or distant metastases and operated tumors with or without node metastases; (b) age—over 50 years and 50 years or younger; (c) treatment during the 3 months before entering the study—radiotherapy or chemotherapy or no treatment; and (d) prognosis—favorable prognosis or unfavorable prognosis. Prognosis was related to frequency and location of metastases. The criteria for favorable prognosis included inoperable disease confined to the breast, or limited to regional or distant lymph nodes, lung or bones. Patients with liver or other visceral metastases, or several metastatic sites, were defined as having unfavorable prognosis. Prognosis was only applied as a matching variable in cases with metastatic disease. Table 1 presents the outcome of the matching procedure.

Table 1

The outcome of the matching procedure on admission (n = 60)

Matching criteria	ABCW n = 60	CBCW n = 60
Stage of disease		
Advancedi	42	42
Limited	18	18
Age		
Above 50 years	20	20
50 years or younger	40	40
Mean/SD	49.35/9.3	49.05/8.5
Range	28–75	29–70
Interval 26–35	3	3
Interval 36–45	19	16
Interval 46–55	23	27
Interval 56–65	10	11
Interval 65–75	5	3
Treatment 3 months before admission		
Treatment	41	41
No treatment	19	19
Prognosis		
Favorable	35	35
Unfavorable	25	25

Abbreviations: ABCW = women with breast cancer, anthroposophic therapy; CBCW = women with breast cancer, conventional therapy.

Sixty-seven percent of the women (n = 40) in the ABCW group and 78% (n = 47) in the CBCW group were married or lived with a partner, and 75% (n = 45) in the ABCW group and 85% (n = 51) in the CBCW group had children. There were no significant difference between the two groups in these respects, but there were differences concerning professions. In the ABCW group, 24% of the women were working in the cultural sphere (four journalists, four artists, three librarians, two actresses and one writer), whereas in the CBCW group, no one was working in that area. A comparison between the two groups on admission to the study showed that the women in the ABCW group perceived their quality of life to be lower and were more anxious than the women in the CBCW group (12).

Table 2 shows the frequency of women in the two groups who answered the questionnaires on the different occasions.

In the ABCW group there were eight women and in the CBCW group four who dropped out of the study. The reasons the women in the ABCW group gave were progression of the disease (n = 2), declining further participation after inclusion because they felt it was too demanding to be interviewed and fill out questionnaires another four times (n = 3), taking part in the interview but not filling in the questionnaires (1), and no answer on invitation to take part in a 6-month or 1-year follow up (n = 2). The reasons the women in the CBCW group gave were progression of the disease (n = 3), and for one woman it was dissatisfaction with the care and the staff of the hospital.

Table 2

The frequency of women who answered the EORTC QLQ-C30 and the LSQ on the different occasions

Occasion	ABCW group			CBCW group			
	Participants	Deceased[1]	Dropouts[1]	Participants	Deceased	Dropouts	Matched pairs
Admission	60			60			60
After 4 weeks	53	1	6	58		2	52
After 3 months	51	3	6	55	3	2	48
After 6 months	48	6	6	53	5	2	44
After 1 year	41	11	8	47	9	4	36

Abbreviations: ABCW = women with breast cancer, anthroposophic therapy; CBCW = women with breast cancer, conventional therapy.
[1]Cumulative frequency.

The instruments

The EORTC QLQ-C30, a well-tested international instrument, consists of 30 items. It generates results expressed in six functional scales (PF = physical functioning, RF = role functioning, EF = emotional functioning, CF = cognitive functioning, SF = social functioning, and QL = global quality of life) and nine symptom scales/multi-item or single items (FA = fatigue, NV = nausea and vomiting, PA = pain, DY = dyspnea, SL = sleep disturbance, AP = appetite loss, CO = constipation, DI = diarrhea, and FI = financial impact). In this study, the Swedish edition of version 1.0 of the instrument was used (13). The scores of the functional scales are linearly transformed such that all scales range from 0 to 100, with a higher scale score representing a higher level of functioning. The symptoms scale items are also linearly transformed such that all scales/items range from 0 to 100. Higher scores represent a higher level of symptomatology/problems (14).

The Life Satisfaction Questionnaire (LSQ), a newly developed instrument (9, 15), consists of 34 items to be answered on a 7-point scale (16); for example, "How much have you been troubled by tiredness during the last week?" Responses range from scale point 1 "very much" to 7 "not at all." Scale point 7 represents the highest quality in all items. The 34 items are allocated to six different factors: physical symptoms (PS), sickness impact (SI), quality of everyday activities (QDA), socioeconomic situation (SES), quality of family relation (QFA), and quality of close friend relationship (QFR). The scores for each factor are calculated in the following way: the raw scores of the items are summarized, divided by the highest point in the scale and multiplied by 100. Thus, 100 represents the maximum quality of life in each factor. The total quality of life is calculated in the same way (16).

The intervention

The women in the ABCW group were treated with a special care program during their stay at the anthroposophic hospital (mean value 14.3 days, range 5–29 days). The individually composed therapies consisted of holistic care including natural products, Iscador, diets, art therapy, eurhythmic therapy, therapeutic massage, and hydrotherapy (6, 12). The women in the other group (CWCB) were treated according to a special regional care program for all patients belonging to the Stockholm/Gotland region consisting of conventional treatment including radiation therapy, chemotherapy or endocrinological therapy (17).

Procedure

The women were followed for 1 year and were assessed by the questionnaires on five occasions: on admission, after 1, 3 and 6 months and after 1 year.

Statistics

An analysis of variance (ANOVA) for repeated measures, paired sample t-tests with 95% confidence intervals and effect sizes (ES) were used for the comparisons. ES was calculated by subtracting the mean difference of the matched pairs in ABCW and CBCW at the posttests (measurement after 1, 3 and 6 months, and 1 year) from the pretest and then dividing by the standard deviation for the pretest (18, 19). A value close to 1 indicates that all of the total variability is attributable to differences between the matched pairs, whereas a value close to 0 indicates that little of the total variability is explained. According to Cohen, an ES greater than 0.20 should be considered small, an ES between 0.50 and 0.80 moderate, and an ES greater 0.80 large (19).

Ethical considerations

The main project was approved after ethical investigation by the Regional Research Ethics Committee at the Karolinska Institute, Stockholm, in 1995 and 2001 (Dnr 95-293 and Dnr 01-015, E. Hamrin).

RESULTS

Eleven women in the ABCW group and nine in the CBCW group died during the data collection period (Table 2). Fifty-two of the matched pairs took part on two occasions,

Table 3

The means for the scales of the EORTC QLQ-C30 for the five different occasions. High values are positive in the functional scales, and low values are positive in the symptom scales (ABCW, n = 36; CBCW, n = 36)

Scale		Admission		4 weeks		3 months		6 months		1 year	
		ABCW	CBCW	ABCW	CBCW	ABCW	CBCW	ABCW	CBCW	ABCW	CBCW
Functional	PF	80.0	72.2	79.4	76.0	75.6	76.7	77.2	74.4	79.2	75.5
	RF	58.3	69.8	60.7	62.9	62.5	62.5	63.9	69.1	63.9	66.7
	EF	51.9	70.0	63.5	70.3	65.5	64.8	71.1	68.1	70.0	63.7
	CF	67.6	76.9	74.1	78.7	76.2	72.7	77.8	73.2	78.7	71.3
	SF	58.8	71.8	71.1	73.6	69.9	74.1	73.6	70.4	63.4	64.8
	QL	49.5	59.5	60.3	60.7	59.3	59.5	63.7	60.4	62.5	54.7
Symptom	FA	52.2	41.7	47.8	37.0	48.2	36.1	47.1	40.1	42.9	43.8
	NV	15.3	9.3	13.4	7.2	11.1	6.5	13.0	5.6	8.6	11.1
	PA	36.6	25.5	24.3	26.9	26.9	25.0	23.6	28.2	25.9	25.0
	DY	25.0	33.3	25.9	36.1	25.9	37.0	25.0	29.6	29.6	40.7
	SL	38.9	33.3	31.9	35.2	34.3	40.7	29.1	45.7	28.7	38.0
	AP	17.6	10.2	13.9	7.4	12.0	10.2	15.7	10.2	8.3	19.4
	CO	26.9	15.7	23.2	19.4	17.6	18.5	18.1	12.0	14.8	14.8
	DI	10.2	19.2	8.8	13.9	11.1	12.7	7.4	13.0	3.7	13.9
	FI	46.3	27.8	44.9	33.8	43.5	29.6	46.3	32.4	42.6	28.5

Abbreviations: ABCW = women with breast cancer, anthroposophic therapy; CBCW = women with breast cancer, conventional therapy; PF = physical functioning; RF = role functioning; EF = emotional functioning; CF = cognitive functioning; SF = social functioning; QL = overall quality of life; FA = fatigue; NV = nausea and vomiting; PA = pain; DY = dyspnea; SL = sleep disturbance; AP = appetite loss; CO = constipation; DI = diarrhea; FI = financial impact.

48 on three occasions, 44 on four occasions and 36 on all occasions. The results are calculated from those 72 women who took part on all occasions, but in Fig. 2, the result of the LSQ is presented with the different number of matched pairs at every occasion.

Changes over time calculated with the help of an ANOVA showed significant changes. In the ABCW group, there were significant changes in two of the scales of the EORTC QLQ-C30 and in four of the LSQ, all representing improvements. In the EORTC QLQ-C30 there were significant improvements in EF ($F = 22.873$; $p = 0.000$) and in QL ($F = 4.517$; $p = 0.002$). In the LSQ improvement was seen in PS ($F = 8.059$; $p = 0.007$), SI ($F = 2.891$; $p = 0.024$), QDA ($F = 13.904$; $p = 0.001$) and SES ($F = 6.150$; $p = 0.000$) In the CBCW group there were no significant changes in any of the scales/factors during the 1-year period.

Table 3 shows the means for the different scales of the EORTC QLQ-C30 for the five different occasions, and in Table 4, the means for the LSQ factors for the five different occasions are shown.

The total quality of life for the 36 matched pairs at the five different occasions is shown in Fig. 1, and the outcome of the different number of matched pairs for the different occasions of the factors of the LSQ is shown in Fig. 2.

Tables 5 and 6 show the mean differences between admission and 6 months and 1 year, respectively, confidence interval and p-values for the different scales of the EORTC QLQ-C30 (Table 5) and the LSQ factors (Table 6) (Student's *t*-test). After 6 months there were significant

Table 4

The means for the LSQ factors for the five different occasions. High values are positive (ABCW, n = 36; CBCW, n = 36)

Factor	Admission		4 weeks		3 months		6 months		1 year	
	ABCW	CBCW	ABCW	CBCW	ABCW	CBCW	ABCW	CBCW	ABCW	CBCW
PS	82.0	87.0	86.2	86.6	86.1	86.1	87.3	89.1	88.2	84.3
SI	58.9	65.3	64.0	63.4	64.6	64.6	66.5	63.8	64.6	64.8
QDA	61.6	65.0	65.4	68.8	69.8	61.5	72.5	68.1	70.6	65.6
SES	64.9	72.3	66.9	70.4	68.3	69.5	70.2	69.0	71.9	67.8
QFA	74.3	71.9	73.7	75.0	72.9	72.9	77.1	75.8	75.3	69.4
QFR	76.6	75.6	76.3	75.0	71.7	70.0	74.4	72.9	74.0	71.8

Abbreviations: ABCW = women with breast cancer, anthroposophic therapy; CBCW = women with breast cancer, conventional therapy; PS = physical symptoms; SI = sickness impact; QDA = quality of everyday activities; SES = socioeconomic situation; QFA = quality of family relation; QFR = quality of close friend relationship.

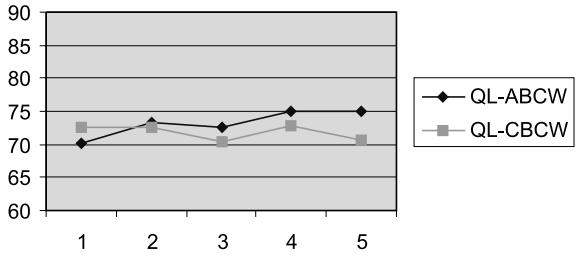


Fig. 1. The total quality of life/life satisfaction measured by the LSQ for the women of the two groups on the five different occasions (n = 36).

differences in three of the functional scales and two of the symptom scales in EORTC QLQ-C30 (Table 5) and at the 1-year follow-up there were differences in three of the functional scales and two of the symptom scales. Concerning measurements with LSQ, the 6-month follow-up (Table 6) showed differences in two of the factors while at the 1-year follow-up there were differences in three of the factors.

Table 7 shows the ES for the differences between admission and the other measurement occasions.

There were some effects due to the difference between the matched pairs. There were small to moderate effects in EF, CF, SF, QL, FA, NV, PA, SL, AP, CO and DI (EORTC QLQ-C30) and in PS, SI, QDA and SES (LSQ) in favor of the ABCW group at the 1-year follow-up.

DISCUSSION

As reported earlier (12), the women in the ABCW group rated their quality of life/life satisfaction lower on entering the study than the women in the CBCW group. The dropout rate was about 13% (n = 8) for the ABCW group and 7% (n = 4) for the CBCW group. The most common reason in the ABCW group was that it was too demanding to take part in the study and in CBCW group it was due to progression of the disease. It is possible that the women in the ABCW group, who had taken the initiative to ask for anthroposophic care, might have been more independent in making decisions, and that could be one explanation for the higher dropout rate. The results of the present study showed clearly, however, that the women in the ABCW group who participated in both the 6-month and the 1-year follow-up

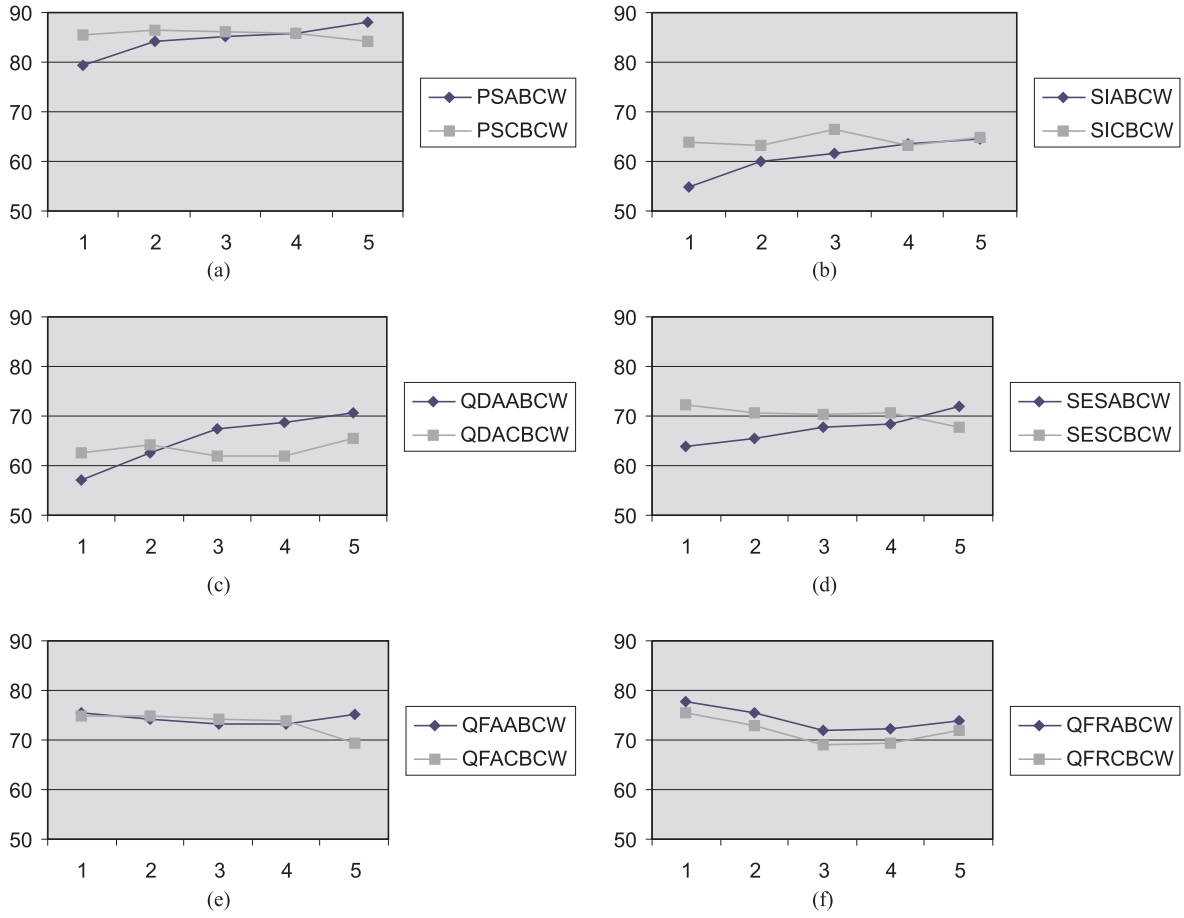


Fig. 2. The quality of life/life satisfaction measured by the six factors of the LSQ for the women of the ABCW and CBCW groups, respectively, on the five different occasions (n = 60, 52, 48, 44, 36). (a) PS = physical symptoms, (b) SI = sickness impact, (c) QDA = quality of daily activities, (d) SES = socioeconomic situation, (e) QFA = quality of family relation, (f) QFR = quality of close friend relationship.

Table 5

The mean difference between admission and 6 months and 1 year, respectively, confidence interval (CI) and p-values for the scales of the EORTC QLQ-C30 (n = 36 matched pairs)[1] (Student's t-test)

Scale	6 months			1 year		
	Mean difference	CI	p	Mean difference	CI	p
PF	−5.0	±11.0	0.363	−4.2	±9.7	0.388
RF	5.9	±17.6	0.501	8.7	±17.4	0.319
EF	21.1	±18.5	0.005	23.4	±10.2	0.000
CF	13.9	±13.9	0.041	16.6	±12.3	0.009
SF	16.2	±16.2	0.033	14.4	±23.7	0.354
QL	13.1	±13.3	0.052	17.8	±13.9	0.013
FA	−3.5	±14.9	0.84	9.1	±14.0	0.107
NV	1.4	±11.4	0.806	−8.1	±9.8	0.101
PA	15.7	±13.2	0.021	10.2	±10.4	0.054
DY	3.7	±13.6	0.593	−2.8	±14.6	0.702
SL	−21.1	±14.2	0.007	−14.8	±14.6	0.047
AP	−1.9	±12.9	0.773	−18.5	±15.1	0.018
CO	−5.1	±17.62	0.561	−11.1	±19.3	0.249
DI	3.4	±11.8	0.431	1.2	±11.9	0.124
FI	−4.6	±10.8	0.392	−4.41	±12.7	0.483

[1]Compare with the values in Table 3.

reported an increased quality of life/life satisfaction after treatment at the anthroposophic hospital on the different occasions up to 1 year after admission to the study in comparison with their matched "twins."

The largest differences in the ABCW group were between admission and the data collection that took place after 1 year. Although the intervention took place between admission and the first data collection occasion after 4 weeks, it is of interest to note that the ratings of different aspects of quality of life/life satisfaction continuously grew higher for the ABCW but not for the CBCW group.

Knobf and Pasacreta (20) maintained that the use of alternative medicine could represent attempts to alleviate symptoms of distress or attempts to retain personal control. A phenomenon that might explain the differences between the groups is that the response shift was higher in the ABCW than the CBCW group. The women in the ABCW group were treated with anthroposophic therapies, where the aims were to mature and to develop spiritual wisdom (6).

Recent reviews have been carried out concerning complementary and alternative medicine (CAM) in relation to cancer (21). However, no randomized controlled study comparing alternative/complementary care and conventional medicine in cancer care has been reported from 1993 onwards (the Cochrane Library Data Base). According to Sprangers (22), the response shift phenomenon might lead to problems in interpretation of assessments completed over time. Randomization could be one way to try to handle that problem. The present study has indeed been a scientific challenge methodologically as strict randomization was not possible (7). This was due to financial reasons and because of the way the health care system functions in Sweden, where the individual patient has to take the initiative to be referred to an anthroposophic clinic. Therefore, our only choice was matching procedures.

Table 6

The mean difference between admission and 6 months and 1 year, respectively, confidence interval (CI) and p-values for the factors of the LSQ (n = 36 matched pairs)[1]

Factor	6 months			1 year		
	Mean difference	CI	P	Mean difference	CI	p
PS	3.1	±6.8	0.368	8.7	±6.2	0.007
SI	9.1	±7.8	0.022	6.2	±7.7	0.109
QDA	7.9	±9.4	0.098	8.5	±7.8	0.033
SES	8.6	±5.4	0.003	11.6	±5.3	0.000
QFA	−1.1	±6.6	0.742	3.5	±6.3	0.274
QFR	0.6	±7.6	0.883	1.3	±5.5	0.647

[1]Compare with the values in Table 4.

Table 7

The effect sizes (ES)[1] for the comparison of the scales of the EORTC QLQ-C30 and the LSQ factors

Scales/factor		4 weeks	3 months	6 months	1 year
EORTC	PF	−0.14	−0.30	−0.17	−0.14
Functional	RF	0.18	0.23	0.12	0.17
	EF	0.30	0.51	0.56	0.71
	CF	0.12	0.30	0.36	0.44
	SF	0.25	0.22	0.41	0.28
	QL	0.26	0.27	0.36	0.49
Symptom	FA	0.01	0.02	−0.04	−0.29
	NV	0.01	−0.05	0.05	0.30
	PA	−0.36	−0.25	−0.42	−0.26
	DY	−0.05	−0.08	0.10	−0.08
	SL	−0.22	−0.31	−0.53	−0.38
	AP	−0.03	−0.16	−0.05	−0.55
	CO	−0.16	−0.27	−0.11	−0.25
	DI	−0.13	−0.03	0.46	0.29
	FI	−0.15	−0.09	−0.09	0.09
LSQ	PS	0.24	0.26	0.16	0.47
	SI	0.27	0.25	0.35	0.24
	QDA	−0.05	0.43	0.29	0.31
	SES	0.14	0.28	0.39	0.52
	QFA	−0.15	−0.09	−0.04	0.14
	QFR	0.02	0.05	0.03	0.08

[1]ES = 0.20–0.50 small; ES = 0.50–0.80 moderate; ES > 0.80 large (19).

The EORTC QLQ-C30 is well tested regarding reliability and validity, internationally as well as in Sweden, with available reference values (23). The LSQ was originally tested concerning reliability and validity in a population of 362 women with breast cancer (9). The explorative factor structure was confirmed through confirmatory factor analyses in a sample of 520 women and men randomly chosen from a Swedish population register (15). The instrument, however, should be further tested with other groups of cancer patients. Both the EORTC QLQ-C30 and the LSQ seemed to fulfill the criteria of responsiveness to change, which is an important criterion in longitudinal validation processes (24). The statistical procedures in the present study followed the recommendations to use effect size to measure change over time as well as more conventional statistical methods. Norman et al. (25) demonstrated that effect size and a minimal clinical important difference (MCID) showed a linear relationship in the measurement of health-related quality of life.

In our study a randomized clinical design would have been preferable, but this was not possible in the health care system where the study took place. Based on the design with matched procedures according to stage of disease, age, earlier treatment and prognosis and on the methodology used in the present study, it is concluded that the women who had chosen anthroposophic care seemed to have increased their perceived quality of life/life satisfaction, which was not found in the matched group.

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